ECA56 Advanced Wireless Communication Systems

CO1 Integrated Engineering Analysis and Planning Assignment

Engineering Design Report: Evolution, Cellular Planning, Spectrum Use and Interference Management for a Regional Mobile Operator

Item	Details
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Course name	5G/6G COMMUNICATION
Course code	ECA5604
Assessment	Open-ended engineering design report (CO1: K4 Analyze, K5 Evaluate, K6 Create)
Prepared for	Network planning team, regional wireless operator

Executive Summary

This report analyzes the operator's planning data in six linked stages (evolution, cell geometry, traffic, mobility, interference and spectrum, coordination and recommendation). Each stage states its requirement, method, calculation and engineering interpretation. All numerical results were computed with a short Python script (Appendix A) and checked by hand. The key findings are:

Quantity	Result	Meaning
λ at 800 MHz; ideal 30 kHz channels in 10 MHz	0.375 m; 333 channels	Idealized: guard bands and duplexing ignored
Cell area; 19-cell cluster area	10.39 km²; 197.45 km²	Hexagonal model, r = 2 km
Reuse fraction; reuse distance (N = 7)	1/7 = 0.143; D = 9.17 km	D/r = 4.58
Offered traffic; blocking at C = 100	80 Erlang; 0.40 %	Meets the 2 % target
Smallest C for ≤ 2 % blocking	92 channels (1.92 %)	One 100-channel group is enough
Overlap distance; hard handoffs per minute	100 m; 1.0 per minute	1 km taken as path length between handoffs
Drop risk per handoff	≈ 1 % (single attempt)	Upper-bound style estimate
C/I (co-channel only); SINR (total)	15.00 dB; 13.81 dB	Target 15 dB **not met** (short by 1.19 dB)
FSPL at 100 m: 900 MHz / 3.5 GHz / 28 GHz	71.53 / 83.32 / 101.38 dB	+11.80 dB and +29.86 dB vs 900 MHz

Recommendation in one line: a four-phase plan. Phase 1 builds a 900 MHz macro coverage layer with 120° sectoring, which lifts SINR to about 16.9 dB. Phase 2 adds a 3.5 GHz mid-band capacity layer on the same sites. Phase 3 adds 28 GHz small cells only in measured hotspots, anchored to the lower bands for mobility. Phase 4 prepares for 6G features (AI-native operation, ISAC). Each step is gated by field measurements of the assumptions listed in Section 8.

1. Requirements, Inputs and Global Assumptions

The planning team needs a defensible network plan from the supplied data. The inputs below are used exactly as given; the general assumptions apply throughout and are repeated where they matter.

Topic	Supplied data
Cell geometry	Hexagonal cells, r = 2 km; planning cluster of 19 cells; reuse cluster size N = 7
Traffic	1,000 subscribers × 0.08 Erlang (busy hour); initial base station C = 100 channels; blocking ≤ 2 %
Mobility	v = 60 km/h; overlap dwell 6 s; route cell length 1 km; handoff interruption 0.5 s; target cell unavailable 1 % of the time
Interference	S = −70 dBm; I_1 (co-channel) = −85 dBm; I_2 (adjacent-channel) = −90 dBm; SINR target 15 dB
Spectrum	900 MHz, 3.5 GHz, 28 GHz at d = 100 m; c = 3×10^8 m/s; FSPL = $20 \log_{10}(4\pi d/\lambda)$
Generations	AMPS (800 MHz, 30 kHz, 10 MHz); GSM (900 MHz, 200 kHz, 8 slots); UMTS (5 MHz); LTE (OFDMA, MIMO); 5G NR (eMBB/URLLC/mMTC, mid-band and mmWave, beamforming); 6G vision (THz, AI-native, holographic, tactile internet, ISAC)

Global assumptions. (i) Flat terrain and identical, regularly spaced sites for the hexagonal model. (ii) Traffic is spatially uniform and stationary in the busy hour. (iii) Erlang B assumptions hold: Poisson arrivals, exponential holding times, blocked calls cleared. (iv) Free-space propagation is used only for the band comparison, as the supplied formula requires; it is a best-case lower bound on loss. (v) The thermal noise floor is not supplied, so SINR is first computed from interference only (noise is discussed in Stage 5).

2. Stage 1: Evolution of Mobile Systems (1G to the 6G Vision)

2.1 Generational comparison

Gen.	Example / spectrum	Main service	Access / radio technology	Key limitation → change it motivated
1G	AMPS, 800 MHz, 30 kHz channels, 10 MHz total	Analog voice only	Analog FM, FDMA (one user per channel)	Low spectral efficiency, needs high C/I (≈ 18 dB for N = 7), easy eavesdropping and cloning, no data → **digital transmission**
2G	GSM, 900 MHz, 200 kHz carrier, 8 slots	Digital voice, SMS, encryption, roaming, low-rate circuit data	FDMA + TDMA (8 users per carrier)	Narrowband circuit-switched design limits data rates → **wideband, packet-based data**
3G	UMTS, 5 MHz carrier	Voice plus mobile Internet and multimedia	WCDMA spread spectrum; reuse-1; soft handoff	Interference-limited, tight power control, rates short of video demand → **OFDMA, all-IP**
4G	LTE	All-IP mobile broadband	OFDMA (downlink), MIMO spatial multiplexing	Sub-6 GHz spectrum scarcity; latency and reliability not designed for critical or massive-IoT uses → **service slicing, new bands**
5G	NR; eMBB, URLLC, massive IoT; mid-band and mmWave	Gb/s broadband, ultra-reliable low latency, massive device density	Flexible OFDM, massive MIMO, beamforming	mmWave blockage and dense-site cost, energy use, no native sensing or AI → **integrated sensing and AI**
6G (vision)	THz and sub-THz; vision under ITU IMT-2030 [2]	Holographic communication, tactile Internet, AI-native networks, ISAC	Not yet standardized	Open issues: THz propagation, hardware, energy, standardization

2.2 Required calculations

Wavelength at 800 MHz:

$$\lambda = c / f = (3 \times 10^8 \text{ m/s}) / (800 \times 10^6 \text{ Hz}) = 0.375 \text{ m}$$

Idealized number of 30 kHz channels in 10 MHz:

$$N(ch) = 10 \text{ MHz} / 30 \text{ kHz} = 333.3 \approx 333 \text{ channels}$$

Practical qualification. This idealized figure explicitly **ignores guard bands and duplexing**. In practice: (a) guard bands at the allocation edges and between operators reduce usable channels; (b) FDD cellular needs paired uplink and downlink, so if the 10 MHz is the total of both directions only about 166 duplex (two-way) channels exist; (c) some channels carry control signalling; and (d) the 333 channels are for the whole system, because with N = 7 each cell can use only about 333/7 ≈ 47 of them.

2G versus 1G. For the same 10 MHz, GSM gives 10 MHz / 200 kHz = 50 carriers × 8 slots = 400 traffic slots (idealized), about 20 % more than AMPS's 333 channels, i.e. 25 kHz per slot against 30 kHz. The larger gains come from digital transmission: speech coding (and later half-rate coding), error correction and equalization tolerate a lower C/I (typically ≈ 9–12 dB against ≈ 18 dB for AMPS) [3][8], which allows tighter reuse. For example N = 4 instead of 7 would give about 400/4 = 100 slots per cell against about 47 channels. Digital signalling also gave encryption, SMS, reliable handoff control and international roaming.

2.3 Synthesis

Interpretation. Across the generations the **role of spectrum** shifted from a scarce resource dedicated one-user-per-channel (1G) to one that is shared by multiplexing (TDMA, CDMA, OFDMA), reused geographically, and extended into wider and higher bands (3.5 GHz, mmWave, THz). **Capacity** growth moved from channelization (1G/2G), to wider carriers and processing gain (3G), to spectral efficiency through MIMO (4G), and finally to spatial reuse through beamforming and densification (5G/6G). **Interference management** moved from planned avoidance (reuse distance and C/I margins in 1G/2G), to averaging and power control (3G), to scheduling, coordination and beam-based spatial separation (4G/5G), and in the 6G vision to AI-driven, sensing-aware management. Every generation therefore made interference a more dynamic, software-controlled problem instead of a fixed frequency-planning one.

3. Stage 2: Cell Coverage and Frequency Reuse

3.1 Geometry and reuse calculations

$$A(\text{cell}) = (3\sqrt{3} / 2)\, r^2 = 2.598 \times (2 \text{ km})^2 = 10.39 \text{ km}^2$$

$$A(\text{cluster}) = 19 \times 10.39 \text{ km}^2 = 197.45 \text{ km}^2$$

$$\text{Reuse fraction} = 1/N = 1/7 = 0.143 \quad (\approx 14.3 \text{ \% of the channels in each cell})$$

$$D = r\sqrt{(3N)} = 2 \text{ km} \times \sqrt{21} = 9.17 \text{ km}; \quad D/r = 4.58$$

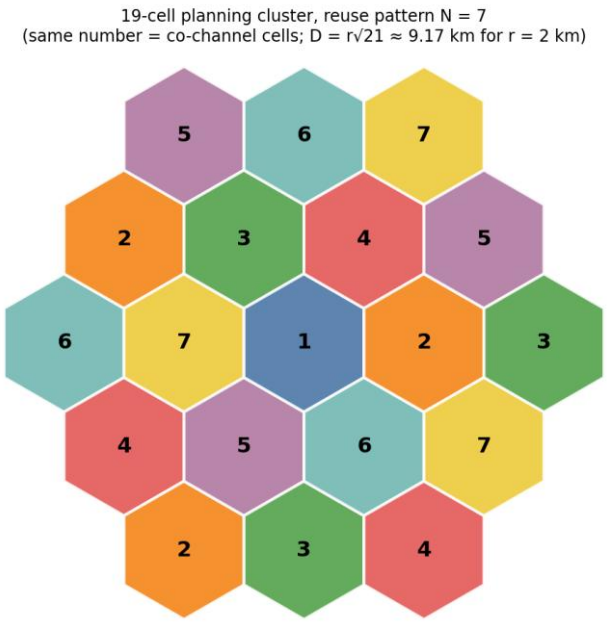


Figure 1. 19-cell hexagonal planning area with N = 7 channel groups (same number = co-channel cells).

Interpretation. Co-channel cells are separated by 9.17 km, 4.58 times the cell radius. A larger D/r gives better C/I but fewer channels per cell. With a path-loss exponent n = 4 and six first-tier interferers, $C/I \approx (D/r)^n / 6 = 441 / 6 = 73.5 \approx 18.7$ dB for an omnidirectional N = 7 layout [3]. This agrees with the classic 18 dB design value for AMPS-type systems.

3.2 Hexagonal approximation: assumptions and limits

- **Assumptions:** flat terrain, identical antennas and powers, uniform path loss in all directions, regular site grid, uniform traffic, and the hexagon as a gap-free stand-in for a circular coverage disc of radius r [5].
- **Geometric coverage is not guaranteed radio coverage.** Real signal strength varies with shadowing (buildings, terrain), multipath fading, indoor penetration, antenna patterns and interference. A point inside a hexagon can still fall below the required signal level, and a point outside can be served. Real cells are irregular and must overlap to allow handoff. Coverage must therefore be defined statistically, for example a required edge-of-cell reliability, and confirmed by drive tests.

3.3 Capacity-expansion techniques

Technique	Effect on coverage	Effect on capacity	Trade-off
Cell splitting	Smaller radius (R/2 → area/4); transmit power reduced (by 2^n, e.g. 16× for n = 4)	Up to ≈ 4× capacity per area	More sites, handoffs roughly double, re-planning
Sectoring (120°, 60°)	Same footprint, directional sectors	Lower C/I problem: first-tier interferers 6 → 2 (≈ +4.8 dB ideal; ≈ 24 dB vs 18.7 dB) so tighter reuse becomes possible	More antennas, intra-site handoffs, trunking loss (channels split per sector)
Microcells	Street-level cells, a few hundred metres	High local capacity in hotspots	Fast users handoff often; needs umbrella cell
Picocells	Indoor or very small area (< 100 m)	Capacity and indoor coverage	Backhaul, cost per site, interference to macro
Femtocells	Home or office, tens of metres, user-installed	Offloads indoor traffic	Uncoordinated co-channel interference, backhaul quality

3.4 Recommendation for the regional operator

The offered traffic is sparse (1,000 subscribers over 197 km², about 5 per km², or about 4.2 Erlang per cell if uniform), so **capacity is not the limiting factor today; coverage and interference are**. The recommended first

measure is therefore **120° sectoring of the macro cells**, not cell splitting. Stage 5 shows it removes the 1.19 dB SINR shortfall. Cell splitting and micro/picocells are held back for measured hotspots as demand grows.

Effects of the recommendation. Interference: the number of first-tier co-channel interferers falls from six to about two (≈ 4.8 dB less co-channel interference, ideal). Handoff: handoff frequency increases because users now also cross sector boundaries (softer or intra-site handoffs); cell splitting by a factor of 2 would double the handoff rate per user because handoff rate ∝ v/R. Trunking: about 4.2 E per cell needs 9 channels unsectored, but 1.4 E per sector needs 5 channels per sector, i.e. 15 per cell (Erlang B at 2 %). With about 100/7 ≈ 14 channels available per cell this is borderline, which the Phase 2 extra carrier (Section 8) resolves.

4. Stage 3: Traffic Capacity (Erlang B)

4.1 Offered traffic

$$A = N(\text{sub}) \times a = 1{,}000 \times 0.08\ \text{E} = 80\ \text{Erlang}$$

Here 0.08 E means each subscriber occupies a channel for 4.8 minutes in the busy hour.

4.2 Erlang B model

The blocking probability for A Erlang offered to C channels is $B(C, A) = (A^C / C!) / \Sigma_{k=0}^{C} (A^k / k!)$. It was evaluated with the numerically stable recursion $B(0) = 1$, $B(k) = A \cdot B(k-1) / (k + A \cdot B(k-1))$ (Appendix A).

Channels C	Blocking B(C, 80)	Carried traffic (E)	Utilization (carried/C)	Meets ≤ 2 %?
80	8.41 %	73.3	91.7 %	No
85	5.05 %	76.0	89.4 %	No
90	2.62 %	77.9	86.6 %	No
91	2.25 %	78.2	85.9 %	No
92	**1.92 %**	**78.5**	**85.3 %**	**Yes (smallest)**
95	1.14 %	79.1	83.3 %	Yes
100	**0.40 %**	**79.7**	**79.7 %**	**Yes**

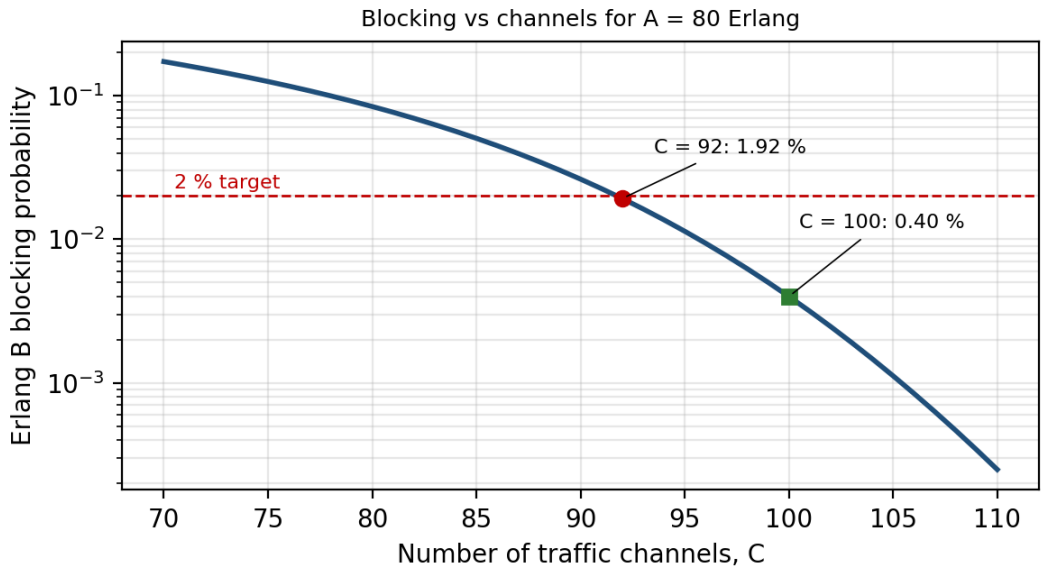


Figure 2. Erlang B blocking versus number of channels for A = 80 Erlang.

4.3 Results

- Blocking with C = 100: **B = 0.40 %**, well below 2 %.
- Smallest channel count meeting 2 %: **C = 92** (B = 1.92 %; C = 91 gives 2.25 %).

- Number of 100-channel groups needed: ⌈92 / 100⌉ = **1 group**, leaving 8 spare channels. At 2 % blocking, 100 channels can carry about 88 E, i.e. about 1,100 subscribers at 0.08 E, which is about 10 % growth headroom.

4.4 Why this is only a first-order estimate

The result treats all 1,000 users as one pooled group. A real plan must also cover: (a) **coverage**: 19 cells (197 km²) need roughly 19 sites regardless of traffic; (b) **spatial split**: if traffic is uniform, each cell sees about 4.21 E and needs 9 channels (blocking 1.7 %), so 19 cells need about 171 channels in total, far more than the pooled 92, a clear trunking-efficiency loss; (c) hotspots and time-of-day peaks; (d) handoff traffic, control channels and guard channels; (e) sectoring, reuse constraints, retries by blocked users, and growth; and (f) backhaul and site engineering.

4.5 Trade-offs and assumptions

Interpretation. Tighter blocking needs more channels with diminishing returns: the 12 channels from 80 to 92 reduce blocking by 6.5 percentage points, while the next 8 (92 to 100) reduce it by only 1.5 points, and utilization falls from 85 % to 80 %. Pooling traffic into big groups is efficient but conflicts with the need for many small cells for coverage. **Assumptions behind the offered-traffic model:** Poisson call arrivals, exponentially distributed holding times, an effectively infinite source population (1,000 ≫ 100), a stationary busy hour, blocked calls cleared (no retries or queueing), all channels equally accessible, and no handoff traffic.

5. Stage 4: Mobility and Handoff

5.1 Distance and handoff rate

$$v = 60 \text{ km/h} = 16.67 \text{ m/s}; \quad d(\text{overlap}) = v \times t = 16.67 \text{ m/s} \times 6 \text{ s} = 100 \text{ m}$$

$$\text{Handoffs per minute} = v / L = (1 \text{ km/min}) / (1 \text{ km}) = 1.0 \text{ per minute}$$

Interpretation of the 1 km length. The supplied "nominal cell length along the route" is taken as the route distance covered in one cell between successive handoff points, with one handoff per cell crossing and no ping-pong. The 100 m overlap is therefore 10 % of that length. If the 1 km were instead only part of a path through a 2 km-radius hexagon (a chord of up to 4 km), the rate would fall to about 0.25 per minute, so the result is sensitive to how the route cuts the cells.

5.2 Hard, soft and softer handoff

Type	How it works	Connection continuity	Resource use
Hard	Break-before-make: old link released, then new link set up; one link at a time (GSM; LTE and NR baseline)	Short interruption (0.5 s here); highest drop risk if the target is unavailable	Lowest: one channel at a time
Soft	Make-before-break: links to two or more cells at once on the same frequency (WCDMA)	No interruption; diversity gain	Higher: extra channels or codes, power and backhaul
Softer	Soft handoff between sectors of the same site; signals combined at the site	No interruption; diversity gain	Extra baseband at the site but no extra backhaul

5.3 Simple call-drop risk

The target cell is unavailable 1 % of the time, so a hard handoff with a single attempt fails with probability $P_1 \approx 0.01$. For a call that experiences k handoffs, assuming independence, $P(\text{drop}, k) = 1 - 0.99^k$. At 1 handoff per minute this gives 1.0 % for a 1-minute call, 3.0 % for 3 minutes, 4.9 % for 5 minutes and 9.6 % for 10 minutes. The 0.5 s interruption is 0.5/60 = 0.83 % of the time at one handoff per minute, and 8.3 % of the 6 s dwell. Since 0.5 s is far shorter than the 6 s dwell, the mobile has about 5.5 s of margin for a retry. Two independent attempts would give $0.01^2 = 10^{-4}$, but this is optimistic.

Interpretation. The estimate is a simple screening number only. It assumes independent handoffs and a fixed 1 % unavailability; it ignores fading and late handoff triggers, ping-pong at the cell edge, speed changes, hysteresis

settings, whether "unavailable" means blocked or radio failure, and the call-duration distribution. The 0.5 s gap also degrades real-time quality (an audible gap) even when the call is not dropped. Practical mitigations are guard channels reserved for handoffs, handoff priority over new calls, and make-before-break (soft handoff or dual-connectivity) techniques.

6. Stage 5: Interference and Spectrum-Band Comparison

6.1 Interference calculation

Carrier-to-interference ratio (co-channel interferer alone):

$$C/I = S - I_1 = -70\ \text{dBm} - (-85\ \text{dBm}) = 15\ \text{dB}$$

Linear summation. Convert dBm to mW with P(mW) = 10^(P_dBm/10):

Quantity	dBm	Linear (mW)	Note
Signal S	−70	1.000×10^{-7}	
Co-channel I_1	−85	3.162×10^{-9}	
Adjacent-channel I_2	−90	1.000×10^{-9}	5 dB below I_1
Total $I = I_1 + I_2$	**−83.81**	**4.162×10^{-9}**	$10 \log_{10}(4.162 \times 10^{-9})$

$$SINR = S - I(\text{total}) = -70 - (-83.81) = 13.81\ \text{dB} < 15\ \text{dB}$$

Interpretation. The 15 dB target is not met; the shortfall is 1.19 dB. The co-channel interferer alone meets the target with exactly zero margin, and the adjacent-channel term (only 31.6 % of the co-channel power) is enough to push the link below it. To comply, total interference must drop to −85 dBm or lower, a reduction of at least 24 % in interference power. A design margin of 2–3 dB for fading would be wiser. No noise figure was supplied, so this is effectively an SIR. As an illustration only, adding a thermal noise floor of −100 dBm (5 MHz bandwidth, 7 dB noise figure) lowers SINR slightly further to about 13.7 dB, so noise does not change the conclusion.

Recommended mitigation: 120° sectoring (with ACLR/guard check)

With 120° sectors the number of first-tier co-channel interferers falls from about six to two, so I_1 falls by about a factor of 3 (4.77 dB, ideal). New values: $I_1' = -89.77$ dBm (1.054×10^{-9} mW), and $I(\text{total})' = 1.054 \times 10^{-9} + 1.0 \times 10^{-9} = 2.054 \times 10^{-9}$ mW = −86.87 dBm, so **$SINR' = -70 + 86.87 = 16.87$ dB**, which meets the target with 1.87 dB of margin. The adjacent-channel term is unchanged, so further gain would need better transmit and receive filtering or a guard channel between neighbouring allocations.

Trade-offs: more antennas and installation cost; trunking-efficiency loss (about 15 channels per cell instead of 9 for the same 2 % blocking, Section 3.4); more intra-site handoffs; and extra planning effort for azimuth and tilt. The alternative of simply raising the cluster size N reduces capacity per cell, and a guard channel wastes spectrum.

6.2 Spectrum-band comparison at d = 100 m

$$\lambda = c/f; \quad FSPL\ (\text{dB}) = 20 \log_{10}(4\pi d/\lambda)$$

Band	λ	FSPL at 100 m	Δ vs 900 MHz	Half-wave dipole	Typical NR carrier / role
900 MHz	33.3 cm	71.53 dB	0 dB	16.7 cm	Up to ~20 MHz; wide-area and indoor coverage
3.5 GHz	8.57 cm	83.32 dB	+11.80 dB	4.29 cm	Up to 100 MHz [7]; capacity layer
28 GHz	1.07 cm	101.38 dB	+29.86 dB	5.4 mm	Up to 400 MHz [7]; hotspot, very high rate

Worked example (900 MHz): $4\pi \times 100 / 0.3333 = 3{,}770$, and $20 \log_{10}(3{,}770) = 71.53$ dB. The differences follow $20 \log_{10}(f_2/f_1)$: $20 \log_{10}(3.5/0.9) = 11.80$ dB and $20 \log_{10}(28/0.9) = 29.86$ dB. For equal free-space loss, range scales as 1/f, so 3.5 GHz reaches about 26 % and 28 GHz about 3 % of the 900 MHz distance.

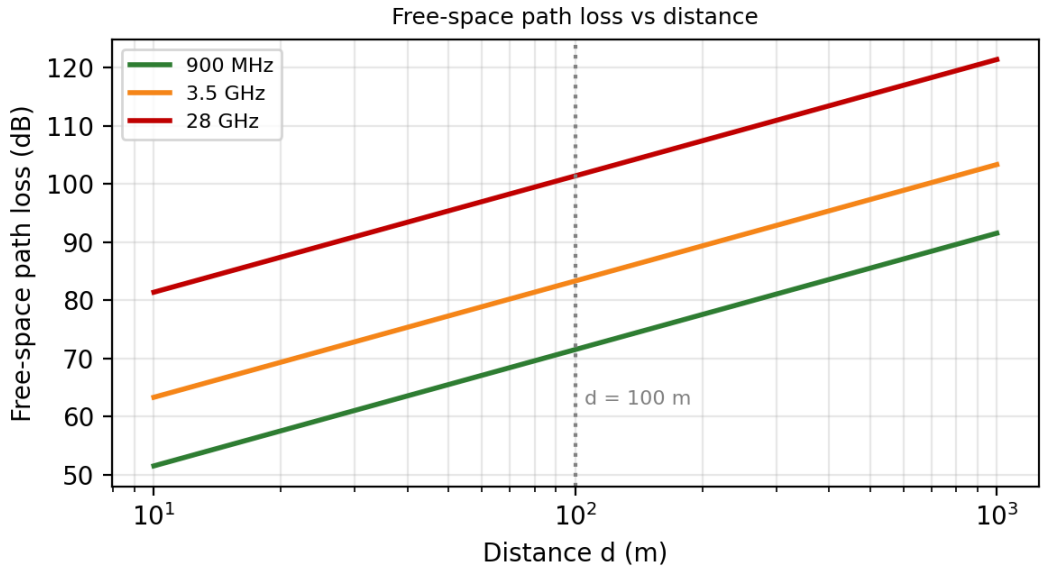


Figure 3. Free-space path loss versus distance for the three bands.

Aspect	900 MHz	3.5 GHz	28 GHz
Penetration	Best: good building and foliage penetration	Moderate: indoor coverage needs more sites or power	Poor: blocked by walls, glass, foliage, body; mostly line-of-sight
Antenna size	Largest, fewer elements per area	Compact; massive MIMO practical	Tiny elements; large arrays in a small area
Coverage	Largest cells (several km)	Medium cells (hundreds of m to ~1–2 km)	Small cells (≈ 100–200 m)
Capacity	Limited by narrow bandwidth	Good: wide carriers plus MIMO	Highest peak: very wide bandwidth, dense spatial reuse

Why higher path loss does not alone decide performance

- **Antenna aperture.** The FSPL formula assumes isotropic antennas, whose effective area ($\lambda^2/4\pi$) shrinks as frequency rises. For a fixed physical aperture, array gain grows as f^2 [10], so a 28 GHz array of the same size can recover up to about 29.9 dB of the extra loss through beamforming, at the cost of beam tracking and alignment.
- **Bandwidth.** Capacity $C = B \log_2(1 + SINR)$ [4]. For example, 20 MHz at 13.8 dB SINR gives about 93 Mb/s, while 400 MHz even at 0 dB SINR gives about 400 Mb/s (single-stream bound).
- **Interference and reuse.** Narrow beams and blockage isolate cells, so mmWave networks are often noise-limited and reuse spectrum aggressively, whereas low-band macro networks are interference-limited.
- **Link budget and deployment:** transmit power, noise figure, fading and penetration margins, line-of-sight probability, site density and cost per bit all matter. FSPL at 100 m is also a best-case lower bound; real urban paths have exponents above 2.

7. Stage 6: Spectrum Coordination and Deployment Recommendation

7.1 Roles in international spectrum coordination

Body	Distinct role
ITU-R	The radiocommunication sector of the International Telecommunication Union (a UN agency). Its study groups carry out technical studies on sharing and compatibility and issue Recommendations and Reports, including the IMT-2020 and IMT-2030 visions [1][2]. It provides the technical basis for global spectrum and satellite-orbit management.

Body	Distinct role
WRC	The World Radiocommunication Conference, held roughly every 3–4 years, reviews and revises the **Radio Regulations**, the binding international treaty [6]. It decides which services may use which bands and identifies bands for IMT (for example 3.4–3.8 GHz portions and, at WRC-19, 24.25–27.5 GHz and 37–43.5 GHz).
BR	The Radiocommunication Bureau is the executive arm of ITU-R, led by an elected Director. It applies the Radio Regulations administratively: it processes frequency-assignment and satellite-network notifications, records them in the Master International Frequency Register, supports coordination between administrations, assists with harmful-interference cases, and supports the study groups and conferences.

Effect on national regulators and operators (one example). Cross-border coordination: a national regulator must align its band plan with the Radio Regulations and agree limits with neighbouring countries. Base stations near borders then face field-strength limits, lower power or guard bands, and co-existence conditions with satellite and other incumbents. For the operator this means licence conditions that restrict power and out-of-band emissions, and that harmonized bands (for example the 3.3–3.8 GHz range) give large device ecosystems, cheaper equipment and easier roaming. Note that to my knowledge the 27.5–29.5 GHz ("28 GHz") range is not part of WRC-19's global IMT identification, so its availability is a national decision that must be confirmed against the current Radio Regulations.

7.2 Integrated, staged deployment strategy

Phase	Actions	Spectrum / technology	Evidence from this report
0. Validate	Drive tests and path-loss measurement; refine traffic forecast; confirm licences and border limits; measure real interference	All candidate bands	Assumptions in Section 8; FSPL and Erlang results are idealized
1. Coverage layer	Macro sites over the 19-cell area; N = 7 reuse; 120° sectors; guard channels for handoff; LTE/NR low-band	900 MHz	Largest range (FSPL 71.5 dB); one 100-channel group covers 80 E at 0.4 %; sectoring lifts SINR 13.81 → 16.87 dB; handoff ≈ 1/min
2. Capacity overlay	Add 3.5 GHz NR with massive MIMO on the same sites; keep 900 MHz for uplink and coverage	3.5 GHz	+11.8 dB loss offset by beamforming; wide carriers; extra carrier relieves the sectoring trunking loss
3. Hotspot densification	28 GHz small cells, micro/picocells in measured hotspots, campuses, fixed wireless; mobility anchored to the lower bands	28 GHz	+29.9 dB loss recovered only with arrays; cells of 100–200 m mean a handoff every 6–12 s at 60 km/h, so a macro anchor is needed
4. 6G readiness	AI-based interference prediction and optimization, ISAC pilots, cloud RAN, sub-THz trials; follow IMT-2030 and WRC outcomes	Existing bands, new bands as allocated	Stage 1 trend: interference management becomes software-driven

Interpretation. The plan follows the evidence in sequence. Coverage and interference, not raw traffic (80 E fits in one 100-channel group), dominate the first stage, so low-band sectored macro cells come first. Capacity growth then comes from added spectrum and spatial reuse (3.5 GHz, then 28 GHz) rather than from splitting every cell, which keeps cost proportional to demand. Handoff results (about 1 per minute, about 1 % drop risk per handoff) show hard handoff is acceptable in the macro layer with guard channels, but fast users should not depend on mmWave cells alone. Each phase is released only when its gate (measured blocking, SINR and utilization) is met.

8. Key Assumptions and Uncertainties to Check Before Deployment

- **Traffic forecast:** 0.08 E per user, 1,000 subscribers, growth, busy-hour peaking and spatial hotspots (uniform traffic is an assumption); whether 100 channels are per site or shared across the N = 7 reuse pool (14 channels per cell on average).
- **Propagation:** real path-loss exponent, shadowing margin, terrain and clutter; FSPL at 100 m is only a best-case comparison, not a link budget.

- **Interference:** the six first-tier co-channel cells combine, so the single −85 dBm interferer is a simplification; load, fading, the real noise floor and antenna patterns; confirm the sectoring gain by measurement.
- **Mobility:** actual route geometry (the 1 km length), speed mix, hysteresis settings, ping-pong, and the real meaning of the 1 % target-cell unavailability.
- **Spectrum and regulation:** availability, bandwidth and licence conditions for 900 MHz, 3.5 GHz and 28 GHz; border coordination; satellite co-existence; device and equipment ecosystem.
- **Cost and energy:** site count, backhaul, power consumption and operating cost per bit for each phase.

9. Conclusion

The analysis shows that the regional network is coverage- and interference-limited today, not traffic-limited: 80 Erlang needs only 92 channels, one 100-channel group (blocking 0.40 %), but the interference case misses its 15 dB target (13.81 dB). Sectoring corrects this to about 16.9 dB. A phased plan of 900 MHz coverage, 3.5 GHz capacity, targeted 28 GHz hotspots and 6G-readiness best matches the evolution trend, the calculated cell and traffic plan, the handoff behaviour and the band characteristics. The conclusions rest on idealized assumptions, so field validation in Phase 0 is essential.

References

[1] ITU-R, Recommendation M.2083-0, “IMT Vision: Framework and overall objectives of the future development of IMT for 2020 and beyond,” 2015.

[2] ITU-R, Recommendation M.2160-0, “Framework and overall objectives of the future development of IMT for 2030 and beyond,” 2023.

[3] T. S. Rappaport, Wireless Communications: Principles and Practice, 2nd ed., Prentice Hall, 2002.

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Note: typical design values quoted (for example AMPS C/I ≈ 18 dB and GSM C/I ≈ 9–12 dB) are standard textbook figures [3][4][8]; the band-plan statements in Section 7 should be re-checked against the current Radio Regulations before use.

Appendix A: Calculation Method (Python)

Inputs: r = 2 km, N = 7, A = 1000 × 0.08, C = 100, S = −70 dBm, I_1 = −85 dBm, I_2 = −90 dBm, f = 0.9, 3.5, 28 GHz, d = 100 m. The script below reproduces every number in this report.

```
import math
c = 3e8
def erlang_b(C, A):           # recursive Erlang B
    B = 1.0
    for k in range(1, C+1):
        B = A*B / (k + A*B)
    return B
A = 1000*0.08                 # 80 Erlang
print(erlang_b(100, A))       # 0.00399
Cmin = next(C for C in range(1,300) if erlang_b(C, A) <= 0.02)   # 92
area = 3*math.sqrt(3)/2*2**2     # 10.39 km^2;  x19 = 197.45 km^2
D = 2*math.sqrt(3*7)          # 9.165 km
I = 10**(-85/10) + 10**(-90/10)  # mW
I_dBm = 10*math.log10(I)      # -83.81
SINR = -70 - I_dBm            # 13.81 dB
for f in (0.9e9, 3.5e9, 28e9):
    lam = c/f
    print(lam, 20*math.log10(4*math.pi*100/lam))   # 71.53, 83.32, 101.38 dB
```